Lab Sheet 2 - Solution

DML Commands - INSERT, UPDATE and DELETE

Objective: To practice Data Manipulation Language (DML) commands - adding, changing and removing rows in tables using SQL (MySQL).

Note: Every query was executed and the output shown is the real result. Each task is a separate query, followed by a SELECT that displays the effect, so you can take a screenshot of each step. Tasks run in order, so later tasks build on earlier data (for example, Q17 uses the Meerut students inserted in Q1 to Q3).

Setup (run first)

Create the database

SQL

```
DROP DATABASE IF EXISTS CollegeDB;
CREATE DATABASE CollegeDB;
```

Output

```
Query OK, 1 row affected

Query OK, 1 row affected
```

Create the tables used in this lab

SQL

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name       VARCHAR(50) NOT NULL,
    Age        INT,
    Course     VARCHAR(30),
    City       VARCHAR(30)
);
CREATE TABLE Employee (
    Emp_ID     INT PRIMARY KEY,
    Emp_Name   VARCHAR(50),
    Department VARCHAR(30),
    Salary     DECIMAL(10,2)
);
CREATE TABLE Department (
    Dept_ID   INT PRIMARY KEY,
    Dept_Name VARCHAR(50),
    Location  VARCHAR(50)
);
CREATE TABLE Product (
    Product_ID   INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Price        DECIMAL(10,2),
    Quantity     INT
);
CREATE TABLE Customer (
    Customer_ID   INT PRIMARY KEY,
    Customer_Name VARCHAR(50),
    Email         VARCHAR(80),
    Phone         VARCHAR(15)
);
CREATE TABLE Course (
    Course_ID   INT PRIMARY KEY,
    Course_Name VARCHAR(50),
    Duration    VARCHAR(20),
    Fees        DECIMAL(10,2)
);
SHOW TABLES;
```

Output

```
Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

+---------------------+
| Tables_in_CollegeDB |
+---------------------+
| Course              |
| Customer            |
| Department          |
| Employee            |
| Product             |
| Student             |
+---------------------+
6 rows in set
```

A. INSERT

Q1. Insert one record into the Student table.

SQL

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (101, 'Rahul Sharma', 20, 'B.Tech', 'Meerut');
SELECT * FROM Student;
```

Output

```
Query OK, 1 row affected

+------------+--------------+------+--------+--------+
| Student_ID | Name         | Age  | Course | City   |
+------------+--------------+------+--------+--------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut |
+------------+--------------+------+--------+--------+
1 row in set
```

Q2. Insert five student records using separate INSERT statements.

SQL

```
INSERT INTO Student VALUES (102, 'Priya Singh', 21, 'BCA', 'Delhi');
INSERT INTO Student VALUES (103, 'Amit Verma', 22, 'MCA', 'Roorkee');
INSERT INTO Student VALUES (104, 'Neha Gupta', 19, 'B.Sc', 'Meerut');
INSERT INTO Student VALUES (105, 'Karan Mehta', 23, 'B.Tech', 'Dehradun');
INSERT INTO Student VALUES (106, 'Sneha Joshi', 20, 'BBA', 'Haridwar');
SELECT * FROM Student;
```

Output

```
Query OK, 1 row affected

Query OK, 1 row affected

Query OK, 1 row affected

Query OK, 1 row affected

Query OK, 1 row affected

+------------+--------------+------+--------+----------+
| Student_ID | Name         | Age  | Course | City     |
+------------+--------------+------+--------+----------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut   |
|        102 | Priya Singh  |   21 | BCA    | Delhi    |
|        103 | Amit Verma   |   22 | MCA    | Roorkee  |
|        104 | Neha Gupta   |   19 | B.Sc   | Meerut   |
|        105 | Karan Mehta  |   23 | B.Tech | Dehradun |
|        106 | Sneha Joshi  |   20 | BBA    | Haridwar |
+------------+--------------+------+--------+----------+
6 rows in set
```

Q3. Insert multiple records into the Student table using a single statement.

SQL

```
INSERT INTO Student VALUES
    (107, 'Vikas Kumar', 21, 'B.Tech', 'Meerut'),
    (108, 'Anjali Rao', 22, 'MBA', 'Delhi'),
    (109, 'Rohit Jain', 20, 'BCA', 'Noida');
SELECT * FROM Student;
```

Output

```
Query OK, 3 rows affected
Records: 3  Duplicates: 0  Warnings: 0

+------------+--------------+------+--------+----------+
| Student_ID | Name         | Age  | Course | City     |
+------------+--------------+------+--------+----------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut   |
|        102 | Priya Singh  |   21 | BCA    | Delhi    |
|        103 | Amit Verma   |   22 | MCA    | Roorkee  |
|        104 | Neha Gupta   |   19 | B.Sc   | Meerut   |
|        105 | Karan Mehta  |   23 | B.Tech | Dehradun |
|        106 | Sneha Joshi  |   20 | BBA    | Haridwar |
|        107 | Vikas Kumar  |   21 | B.Tech | Meerut   |
|        108 | Anjali Rao   |   22 | MBA    | Delhi    |
|        109 | Rohit Jain   |   20 | BCA    | Noida    |
+------------+--------------+------+--------+----------+
9 rows in set
```

Q4. Insert records into the Employee table.

SQL

```
INSERT INTO Employee VALUES
    (1, 'Arun Kapoor', 'IT', 55000),
    (2, 'Meena Iyer', 'HR', 42000),
    (3, 'Suresh Patel', 'Finance', 48000),
    (4, 'Divya Menon', 'IT', 61000),
    (5, 'Ravi Das', 'Sales', 30000);
SELECT * FROM Employee;
```

Output

```
Query OK, 5 rows affected
Records: 5  Duplicates: 0  Warnings: 0

+--------+--------------+------------+----------+
| Emp_ID | Emp_Name     | Department | Salary   |
+--------+--------------+------------+----------+
|      1 | Arun Kapoor  | IT         | 55000.00 |
|      2 | Meena Iyer   | HR         | 42000.00 |
|      3 | Suresh Patel | Finance    | 48000.00 |
|      4 | Divya Menon  | IT         | 61000.00 |
|      5 | Ravi Das     | Sales      | 30000.00 |
+--------+--------------+------------+----------+
5 rows in set
```

Q5. Insert records into the Department table.

SQL

```
INSERT INTO Department VALUES
    (1, 'Computer Science', 'Block A'),
    (2, 'Mathematics', 'Block B'),
    (3, 'Physics', 'Block C'),
    (4, 'Commerce', 'Block D');
SELECT * FROM Department;
```

Output

```
Query OK, 4 rows affected
Records: 4  Duplicates: 0  Warnings: 0

+---------+------------------+----------+
| Dept_ID | Dept_Name        | Location |
+---------+------------------+----------+
|       1 | Computer Science | Block A  |
|       2 | Mathematics      | Block B  |
|       3 | Physics          | Block C  |
|       4 | Commerce         | Block D  |
+---------+------------------+----------+
4 rows in set
```

Q6. Insert records into the Product table.

SQL

```
INSERT INTO Product VALUES
    (1, 'Laptop', 55000, 10),
    (2, 'Mouse', 500, 50),
    (3, 'Keyboard', 1200, 0),
    (4, 'Monitor', 9000, 15),
    (5, 'Webcam', 2500, 0);
SELECT * FROM Product;
```

Output

```
Query OK, 5 rows affected
Records: 5  Duplicates: 0  Warnings: 0

+------------+--------------+----------+----------+
| Product_ID | Product_Name | Price    | Quantity |
+------------+--------------+----------+----------+
|          1 | Laptop       | 55000.00 |       10 |
|          2 | Mouse        |   500.00 |       50 |
|          3 | Keyboard     |  1200.00 |        0 |
|          4 | Monitor      |  9000.00 |       15 |
|          5 | Webcam       |  2500.00 |        0 |
+------------+--------------+----------+----------+
5 rows in set
```

Q7. Insert records into the Customer table.

SQL

```
INSERT INTO Customer VALUES
    (1, 'Anil Khanna', 'anil@mail.com', '9876543210'),
    (2, 'Sunita Rao', 'sunita@mail.com', '9123456780'),
    (3, 'Manoj Tiwari', 'manoj@mail.com', '9988776655');
SELECT * FROM Customer;
```

Output

```
Query OK, 3 rows affected
Records: 3  Duplicates: 0  Warnings: 0

+-------------+---------------+-----------------+------------+
| Customer_ID | Customer_Name | Email           | Phone      |
+-------------+---------------+-----------------+------------+
|           1 | Anil Khanna   | anil@mail.com   | 9876543210 |
|           2 | Sunita Rao    | sunita@mail.com | 9123456780 |
|           3 | Manoj Tiwari  | manoj@mail.com  | 9988776655 |
+-------------+---------------+-----------------+------------+
3 rows in set
```

Q8. Insert records into the Course table.

SQL

```
INSERT INTO Course VALUES
    (1, 'Data Structures', '6 Months', 15000),
    (2, 'DBMS', '4 Months', 12000),
    (3, 'Python', '3 Months', 8000),
    (4, 'Machine Learning', '8 Months', 25000);
SELECT * FROM Course;
```

Output

```
Query OK, 4 rows affected
Records: 4  Duplicates: 0  Warnings: 0

+-----------+------------------+----------+----------+
| Course_ID | Course_Name      | Duration | Fees     |
+-----------+------------------+----------+----------+
|         1 | Data Structures  | 6 Months | 15000.00 |
|         2 | DBMS             | 4 Months | 12000.00 |
|         3 | Python           | 3 Months |  8000.00 |
|         4 | Machine Learning | 8 Months | 25000.00 |
+-----------+------------------+----------+----------+
4 rows in set
```

Q9. Insert a student record with values for all columns.

SQL

```
INSERT INTO Student VALUES (110, 'Pooja Nair', 19, 'B.Sc', 'Roorkee');
SELECT * FROM Student WHERE Student_ID = 110;
```

Output

```
Query OK, 1 row affected

+------------+------------+------+--------+---------+
| Student_ID | Name       | Age  | Course | City    |
+------------+------------+------+--------+---------+
|        110 | Pooja Nair |   19 | B.Sc   | Roorkee |
+------------+------------+------+--------+---------+
1 row in set
```

Q10. Insert a student record by specifying only selected columns.

SQL

```
-- Age, Course and City are left out, so they become NULL
INSERT INTO Student (Student_ID, Name) VALUES (111, 'Deepak Yadav');
SELECT * FROM Student WHERE Student_ID = 111;
```

Output

```
Query OK, 1 row affected

+------------+--------------+------+--------+------+
| Student_ID | Name         | Age  | Course | City |
+------------+--------------+------+--------+------+
|        111 | Deepak Yadav | NULL | NULL   | NULL |
+------------+--------------+------+--------+------+
1 row in set
```

B. UPDATE

Q11. Update the city of a particular student.

SQL

```
UPDATE Student SET City = 'Lucknow' WHERE Student_ID = 102;
SELECT Student_ID, Name, City FROM Student WHERE Student_ID = 102;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+------------+-------------+---------+
| Student_ID | Name        | City    |
+------------+-------------+---------+
|        102 | Priya Singh | Lucknow |
+------------+-------------+---------+
1 row in set
```

Q12. Update the salary of a particular employee.

SQL

```
UPDATE Employee SET Salary = 58000 WHERE Emp_ID = 1;
SELECT * FROM Employee WHERE Emp_ID = 1;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+--------+-------------+------------+----------+
| Emp_ID | Emp_Name    | Department | Salary   |
+--------+-------------+------------+----------+
|      1 | Arun Kapoor | IT         | 58000.00 |
+--------+-------------+------------+----------+
1 row in set
```

Q13. Increase the salary of all employees by 10%.

SQL

```
UPDATE Employee SET Salary = Salary * 1.10;
SELECT * FROM Employee;
```

Output

```
Query OK, 5 rows affected
Rows matched: 5  Changed: 5  Warnings: 0

+--------+--------------+------------+----------+
| Emp_ID | Emp_Name     | Department | Salary   |
+--------+--------------+------------+----------+
|      1 | Arun Kapoor  | IT         | 63800.00 |
|      2 | Meena Iyer   | HR         | 46200.00 |
|      3 | Suresh Patel | Finance    | 52800.00 |
|      4 | Divya Menon  | IT         | 67100.00 |
|      5 | Ravi Das     | Sales      | 33000.00 |
+--------+--------------+------------+----------+
5 rows in set
```

Q14. Increase the price of all products by 5%.

SQL

```
UPDATE Product SET Price = Price * 1.05;
SELECT * FROM Product;
```

Output

```
Query OK, 5 rows affected
Rows matched: 5  Changed: 5  Warnings: 0

+------------+--------------+----------+----------+
| Product_ID | Product_Name | Price    | Quantity |
+------------+--------------+----------+----------+
|          1 | Laptop       | 57750.00 |       10 |
|          2 | Mouse        |   525.00 |       50 |
|          3 | Keyboard     |  1260.00 |        0 |
|          4 | Monitor      |  9450.00 |       15 |
|          5 | Webcam       |  2625.00 |        0 |
+------------+--------------+----------+----------+
5 rows in set
```

Q15. Change the course of a particular student.

SQL

```
UPDATE Student SET Course = 'MBA' WHERE Student_ID = 103;
SELECT Student_ID, Name, Course FROM Student WHERE Student_ID = 103;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+------------+------------+--------+
| Student_ID | Name       | Course |
+------------+------------+--------+
|        103 | Amit Verma | MBA    |
+------------+------------+--------+
1 row in set
```

Q16. Update the department of an employee.

SQL

```
UPDATE Employee SET Department = 'Finance' WHERE Emp_ID = 2;
SELECT Emp_ID, Emp_Name, Department FROM Employee WHERE Emp_ID = 2;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+--------+------------+------------+
| Emp_ID | Emp_Name   | Department |
+--------+------------+------------+
|      2 | Meena Iyer | Finance    |
+--------+------------+------------+
1 row in set
```

Q17. Update the city of all students from Meerut to Delhi.

SQL

```
UPDATE Student SET City = 'Delhi' WHERE City = 'Meerut';
SELECT Student_ID, Name, City FROM Student;
```

Output

```
Query OK, 3 rows affected
Rows matched: 3  Changed: 3  Warnings: 0

+------------+--------------+----------+
| Student_ID | Name         | City     |
+------------+--------------+----------+
|        101 | Rahul Sharma | Delhi    |
|        102 | Priya Singh  | Lucknow  |
|        103 | Amit Verma   | Roorkee  |
|        104 | Neha Gupta   | Delhi    |
|        105 | Karan Mehta  | Dehradun |
|        106 | Sneha Joshi  | Haridwar |
|        107 | Vikas Kumar  | Delhi    |
|        108 | Anjali Rao   | Delhi    |
|        109 | Rohit Jain   | Noida    |
|        110 | Pooja Nair   | Roorkee  |
|        111 | Deepak Yadav | NULL     |
+------------+--------------+----------+
11 rows in set
```

Q18. Update the fees of a particular course.

SQL

```
UPDATE Course SET Fees = 13500 WHERE Course_ID = 2;
SELECT * FROM Course WHERE Course_ID = 2;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+-----------+-------------+----------+----------+
| Course_ID | Course_Name | Duration | Fees     |
+-----------+-------------+----------+----------+
|         2 | DBMS        | 4 Months | 13500.00 |
+-----------+-------------+----------+----------+
1 row in set
```

Q19. Update multiple columns of a student simultaneously.

SQL

```
UPDATE Student
SET Name = 'Rahul Kumar Sharma', Age = 21, Course = 'M.Tech'
WHERE Student_ID = 101;
SELECT * FROM Student WHERE Student_ID = 101;
```

Output

```
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+------------+--------------------+------+--------+-------+
| Student_ID | Name               | Age  | Course | City  |
+------------+--------------------+------+--------+-------+
|        101 | Rahul Kumar Sharma |   21 | M.Tech | Delhi |
+------------+--------------------+------+--------+-------+
1 row in set
```

C. DELETE

Q20. Delete a particular student record using Student_ID.

SQL

```
DELETE FROM Student WHERE Student_ID = 111;
SELECT * FROM Student;
```

Output

```
Query OK, 1 row affected

+------------+--------------------+------+--------+----------+
| Student_ID | Name               | Age  | Course | City     |
+------------+--------------------+------+--------+----------+
|        101 | Rahul Kumar Sharma |   21 | M.Tech | Delhi    |
|        102 | Priya Singh        |   21 | BCA    | Lucknow  |
|        103 | Amit Verma         |   22 | MBA    | Roorkee  |
|        104 | Neha Gupta         |   19 | B.Sc   | Delhi    |
|        105 | Karan Mehta        |   23 | B.Tech | Dehradun |
|        106 | Sneha Joshi        |   20 | BBA    | Haridwar |
|        107 | Vikas Kumar        |   21 | B.Tech | Delhi    |
|        108 | Anjali Rao         |   22 | MBA    | Delhi    |
|        109 | Rohit Jain         |   20 | BCA    | Noida    |
|        110 | Pooja Nair         |   19 | B.Sc   | Roorkee  |
+------------+--------------------+------+--------+----------+
10 rows in set
```

Q21. Delete all students belonging to a particular city.

SQL

```
DELETE FROM Student WHERE City = 'Noida';
SELECT * FROM Student;
```

Output

```
Query OK, 1 row affected

+------------+--------------------+------+--------+----------+
| Student_ID | Name               | Age  | Course | City     |
+------------+--------------------+------+--------+----------+
|        101 | Rahul Kumar Sharma |   21 | M.Tech | Delhi    |
|        102 | Priya Singh        |   21 | BCA    | Lucknow  |
|        103 | Amit Verma         |   22 | MBA    | Roorkee  |
|        104 | Neha Gupta         |   19 | B.Sc   | Delhi    |
|        105 | Karan Mehta        |   23 | B.Tech | Dehradun |
|        106 | Sneha Joshi        |   20 | BBA    | Haridwar |
|        107 | Vikas Kumar        |   21 | B.Tech | Delhi    |
|        108 | Anjali Rao         |   22 | MBA    | Delhi    |
|        110 | Pooja Nair         |   19 | B.Sc   | Roorkee  |
+------------+--------------------+------+--------+----------+
9 rows in set
```

Q22. Delete employees whose salary is below a specified amount.

SQL

```
DELETE FROM Employee WHERE Salary < 35000;
SELECT * FROM Employee;
```

Output

```
Query OK, 1 row affected

+--------+--------------+------------+----------+
| Emp_ID | Emp_Name     | Department | Salary   |
+--------+--------------+------------+----------+
|      1 | Arun Kapoor  | IT         | 63800.00 |
|      2 | Meena Iyer   | Finance    | 46200.00 |
|      3 | Suresh Patel | Finance    | 52800.00 |
|      4 | Divya Menon  | IT         | 67100.00 |
+--------+--------------+------------+----------+
4 rows in set
```

Q23. Delete products whose quantity is zero.

SQL

```
DELETE FROM Product WHERE Quantity = 0;
SELECT * FROM Product;
```

Output

```
Query OK, 2 rows affected

+------------+--------------+----------+----------+
| Product_ID | Product_Name | Price    | Quantity |
+------------+--------------+----------+----------+
|          1 | Laptop       | 57750.00 |       10 |
|          2 | Mouse        |   525.00 |       50 |
|          4 | Monitor      |  9450.00 |       15 |
+------------+--------------+----------+----------+
3 rows in set
```

Q24. Delete records satisfying multiple conditions using AND/OR.

SQL

```
-- delete BBA students younger than 21, OR any student from Dehradun
DELETE FROM Student
WHERE (Course = 'BBA' AND Age < 21) OR City = 'Dehradun';
SELECT * FROM Student;
```

Output

```
Query OK, 2 rows affected

+------------+--------------------+------+--------+---------+
| Student_ID | Name               | Age  | Course | City    |
+------------+--------------------+------+--------+---------+
|        101 | Rahul Kumar Sharma |   21 | M.Tech | Delhi   |
|        102 | Priya Singh        |   21 | BCA    | Lucknow |
|        103 | Amit Verma         |   22 | MBA    | Roorkee |
|        104 | Neha Gupta         |   19 | B.Sc   | Delhi   |
|        107 | Vikas Kumar        |   21 | B.Tech | Delhi   |
|        108 | Anjali Rao         |   22 | MBA    | Delhi   |
|        110 | Pooja Nair         |   19 | B.Sc   | Roorkee |
+------------+--------------------+------+--------+---------+
7 rows in set
```

D. Complete INSERT-UPDATE-DELETE sequence

Q25. Complete sequence of INSERT, UPDATE and DELETE on a Student table (table shown after each operation).

SQL

```
-- a separate demo table so the sequence starts from empty
CREATE TABLE Student_Demo (
    Student_ID INT PRIMARY KEY,
    Name       VARCHAR(50) NOT NULL,
    Age        INT,
    Course     VARCHAR(30),
    City       VARCHAR(30)
);

-- 1) INSERT
INSERT INTO Student_Demo VALUES
    (1, 'Aman Gupta', 20, 'B.Tech', 'Meerut'),
    (2, 'Ritu Sharma', 21, 'BCA', 'Delhi'),
    (3, 'Sahil Khan', 22, 'MCA', 'Noida');
SELECT * FROM Student_Demo;

-- 2) UPDATE
UPDATE Student_Demo SET City = 'Delhi', Age = 21 WHERE Student_ID = 1;
SELECT * FROM Student_Demo;

-- 3) DELETE
DELETE FROM Student_Demo WHERE Student_ID = 3;
SELECT * FROM Student_Demo;
```

Output

```
Query OK, 0 rows affected

Query OK, 3 rows affected
Records: 3  Duplicates: 0  Warnings: 0

+------------+-------------+------+--------+--------+
| Student_ID | Name        | Age  | Course | City   |
+------------+-------------+------+--------+--------+
|          1 | Aman Gupta  |   20 | B.Tech | Meerut |
|          2 | Ritu Sharma |   21 | BCA    | Delhi  |
|          3 | Sahil Khan  |   22 | MCA    | Noida  |
+------------+-------------+------+--------+--------+
3 rows in set

Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0

+------------+-------------+------+--------+-------+
| Student_ID | Name        | Age  | Course | City  |
+------------+-------------+------+--------+-------+
|          1 | Aman Gupta  |   21 | B.Tech | Delhi |
|          2 | Ritu Sharma |   21 | BCA    | Delhi |
|          3 | Sahil Khan  |   22 | MCA    | Noida |
+------------+-------------+------+--------+-------+
3 rows in set

Query OK, 1 row affected

+------------+-------------+------+--------+-------+
| Student_ID | Name        | Age  | Course | City  |
+------------+-------------+------+--------+-------+
|          1 | Aman Gupta  |   21 | B.Tech | Delhi |
|          2 | Ritu Sharma |   21 | BCA    | Delhi |
+------------+-------------+------+--------+-------+
2 rows in set
```